Stress Testing Framework

Measuring Portfolio Resilience Under Extreme Scenarios

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Abstract

An applied, example-driven treatment of stress testing for financial portfolios. We begin from the practical limitations of statistical risk measures (VaR, ES) that are calibrated to "normal" market conditions, and develop the three pillars of stress testing: historical scenario replay, hypothetical (regulatory) scenarios, and reverse stress testing. The Fed DFAST architecture is examined in detail, including the macro-to-portfolio transmission mechanism via regression-based and satellite models. Credit migration under stress and reverse stress testing as an optimization problem are treated with worked numerical examples throughout. The document assumes familiarity with VaR/ES (Project 03), volatility modeling (Project 04), and tail risk (Project 05).

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1 Why Stress Testing

Example 1.1 (The 2008 wake-up call)**.** In the first half of 2007, the $\mathrm{VaR}_{0.99}$ for a diversified portfolio of US equities and mortgage-backed securities was typically estimated at 2–3% of portfolio value (daily). The GARCH models were calibrated to 2003–2007 data — a period of unusually low volatility (the "Great Moderation").

Between September and November 2008, daily portfolio losses of 5–8% occurred on multiple occasions. The 99% VaR was breached not once or twice, but *dozens* of times in a single quarter. Models that were "correct" under recent history were catastrophically wrong under crisis conditions.

The failure was not that VaR or ES were computed incorrectly. The failure was **structural**: statistical risk measures answer the question "what is the worst loss under the distribution estimated from recent data?" They do *not* answer the question "what happens if the world changes in a specific, adverse way?"

Stress testing is the complement to statistical risk measures. It asks a fundamentally different question:

Core Idea

Statistical risk measures (VaR, ES): "Given the current regime, what is the α-quantile or tail expectation of the loss distribution?"

Stress testing: "Given a specific adverse scenario (whether historical or hypothetical), what would the portfolio lose?"

The two approaches are complementary, not competing. VaR/ES measures the tail *within* the estimated distribution. Stress testing evaluates losses *outside* the estimated distribution — under scenarios that the model may assign negligible probability to, but that regulators, risk managers, and history tell us can happen.

The regulatory mandate is explicit and growing:

- **Basel III/IV** (BCBS, 2018): banks must conduct regular stress tests covering credit, market, and operational risk. Internal stress testing is a pillar of the Internal Capital Adequacy Assessment Process (ICAAP).
- **DFAST / CCAR** (US): the Dodd-Frank Act Stress Tests require all bank holding companies with assets above $100B to project losses under Fed-prescribed macroeconomic scenarios.
- **EBA / ECB** (EU): the European Banking Authority conducts biennial EU-wide stress tests with prescribed adverse scenarios.
- **PRA** (UK): the Bank of England's Prudential Regulation Authority runs concurrent stress tests for major UK banks.

Warning

The 2008 crisis did not just reveal that models were miscalibrated. It revealed a **conceptual gap**: risk management frameworks built entirely on statistical measures estimated from recent data will always be blind to regime shifts. Stress testing fills this gap by forcing institutions to evaluate losses under scenarios that are not in the historical sample — or that occurred so long ago that statistical models have "forgotten" them.

2 Types of Stress Tests

Stress tests come in three broad categories, each answering a different question.

2.1 Historical Scenarios

Approach: take the actual market moves observed during a past crisis and apply them to the current portfolio.

Question answered: "If the 2008 GFC happened again today, what would we lose?"

Advantages:

- Scenarios are internally consistent (all factor moves actually co-occurred).
- Easy to communicate to senior management ("this is what happened last time").
- No modeling assumptions needed — just apply the historical factor changes.

Limitations:

- The next crisis may look different from any past crisis.
- The portfolio composition today may have exposures to risk factors that did not exist or were not material in the historical period.
- Historical scenarios are backward-looking by definition.

2.2 Hypothetical Scenarios

Approach: construct a scenario from economic logic or regulatory specification, specifying paths for key macroeconomic variables over a projection horizon.

Question answered: "If GDP fell by 6% and unemployment rose to 10%, what would our portfolio lose?"

Examples: the Fed DFAST baseline, adverse, and severely adverse scenarios; internal scenarios designed by a bank's economics team.

Advantages:

- Forward-looking: can incorporate risks that have not yet materialized.
- Tailored to specific vulnerabilities of the institution.
- Allows exploration of "what if" questions that history has not answered.

Limitations:

- The scenario may be internally inconsistent (e.g., GDP falls but equity markets rise — is that plausible?).
- Requires a transmission model from macro variables to portfolio losses.
- Results are only as good as the transmission model.

2.3 Reverse Stress Testing

Approach: instead of specifying a scenario and computing the loss, specify a loss threshold and find the scenario that produces it.

Question answered: "What scenario would cause our portfolio to lose enough to breach our capital ratio?"

Advantages:

- Identifies hidden vulnerabilities that forward stress tests may miss.
- Forces the institution to think about its "break point."
- Required by several regulators (PRA, EBA).

Limitations:

- Computationally harder (inverse problem).
- Multiple scenarios may produce the same loss — which one is "the" answer?
- Requires careful formulation of the optimization problem.

2.4 Comparison

	Historical	**Hypothetical**	**Reverse**
Input	Past crisis dates	Macro variable paths	Loss threshold
Output	Portfolio loss	Portfolio loss	Scenario
Direction	Scenario → Loss	Scenario → Loss	Loss → Scenario
Key strength	Realism, consistency	Forward-looking	Finds blind spots
Key weakness	Backward-looking	Model-dependent	Computational cost
Regulatory use	Supplementary	DFAST/CCAR, EBA	PRA, ICAAP

Table 1: Comparison of the three stress testing approaches.

Key Insight

A robust stress testing framework uses *all three* approaches. Historical scenarios provide a reality anchor. Hypothetical scenarios explore forward-looking risks. Reverse stress testing ensures no critical vulnerability is overlooked. The three approaches are complementary, not substitutes.

3 Fed DFAST Scenario Architecture

The Federal Reserve's Dodd-Frank Act Stress Tests (DFAST) provide the most structured example of hypothetical stress testing. Each year, the Fed publishes three macroeconomic scenarios, and covered institutions must project their capital ratios under each.

3.1 The Three Scenarios

1. **Baseline**: a "most likely" economic path. Roughly consistent with the Blue Chip consensus forecast. GDP growth $\approx 2\%$, unemployment $\approx 4\%$, equity indices roughly flat.

2. **Adverse**: a moderate recession. GDP growth $\approx -1\%$ to -2%, unemployment rises to $\approx 7\%$, equity indices decline $\approx 25\%$, credit spreads widen moderately.

3. **Severely adverse**: a deep recession. GDP growth $\approx -6\%$ to -8%, unemployment rises to $\approx 10\%$, equity indices decline $\approx 55\%$, credit spreads spike, house prices fall $\approx 25\%$.

3.2 Macro Variable Paths

The Fed specifies quarterly paths for 28 macroeconomic and financial variables over a 9-quarter (or 13-quarter) projection horizon. The key variables include:

Variable	Baseline	Adverse	Severely Adverse	Units
Real GDP growth (Y/Y)	+2.0%	−2.0%	−6.0%	%
Unemployment rate (peak)	4.2%	7.0%	10.0%	%
Dow Jones / S&P trough	−2%	−25%	−55%	% from current
10Y Treasury yield (trough)	3.8%	2.5%	1.0%	%
BBB corporate spread (peak)	+50bps	+200bps	+450bps	bps
House Price Index (HPI)	+2%	−10%	−25%	%
VIX (peak)	20	45	70	index

Table 2: Representative DFAST macro variable paths (stylized from recent exercises). Actual Fed scenarios vary year to year.

3.3 What Banks Must Do

For each scenario, covered institutions must:

1. Project quarterly **pre-provision net revenue** (PPNR).

2. Project quarterly **credit losses** (provision for loan losses).

3. Project **trading and counterparty losses** (for firms with large trading books).

4. Project **operational risk losses**.

5. Compute **capital ratios** (CET1, Tier 1, Total Capital, Leverage) at each quarter of the horizon.

6. Demonstrate that capital ratios stay above regulatory minimums *at every quarter*, not just at the end.

Warning

The DFAST severely adverse scenario is designed to be harsh but *plausible*. It is not a "worst case" — the 2008 crisis was actually more severe on some dimensions (e.g., the S&P 500 fell 57% peak-to-trough, and unemployment reached 10%). The regulatory scenario is a floor, not a ceiling, for stress testing. Institutions are expected to run their own, potentially more severe, internal scenarios.

3.4 Worked Example: Severely Adverse Impact

Example 3.1 (Severely adverse scenario)**.** Consider a bank with a \$50 billion loan portfolio (60% commercial, 40% consumer) and a \$10 billion trading book (equity-heavy).

Under the severely adverse scenario:

- **Credit losses**: the loan portfolio's weighted average PD increases from 1.2% (baseline) to 4.5% (severely adverse), driven by GDP contraction and unemployment. Expected credit losses: $\$50\text{B} \times (4.5\% - 1.2\%) \times 0.45$ (LGD) $\approx \$743\text{M}$.
- **Trading losses**: equity index -55%, credit spreads $+450$bps. Estimated mark-to-market loss on the trading book: $\approx \$2.8\text{B}$.
- **PPNR offset**: even under stress, the bank earns net interest income and fees. Projected 9-quarter cumulative PPNR: $\approx \$1.5\text{B}$.
- **Net stress loss**: $\$743\text{M} + \$2.8\text{B} - \$1.5\text{B} = \$2.04\text{B}$.

If the bank's CET1 capital is \$5.0 billion and risk-weighted assets are \$55 billion, the pre-stress CET1 ratio is $5.0/55 = 9.1\%$. After the stress loss: $(5.0-2.04)/55 = 5.4\%$. The regulatory minimum CET1 ratio is 4.5%, plus a stress capital buffer. The bank survives, but with minimal headroom.

Interpretation

The DFAST exercise is not just a compliance exercise. It reveals the **capital adequacy** of the institution under severe but plausible conditions. A bank that barely passes the severely adverse scenario has little buffer against an even more severe event — and the 2008 crisis was, on several dimensions, more severe than the typical DFAST scenario.

4 Macro-to-Portfolio Transmission

The core modeling challenge in hypothetical stress testing: how do macroeconomic shocks translate into portfolio losses? The scenario gives us ΔGDP, ΔUnemployment, ΔEquity Index, etc. We need to convert these into a dollar loss for the portfolio.

Two main approaches exist.

4.1 Approach 1: Regression-Based (Top-Down)

Core Idea

The regression approach: regress portfolio returns (or P&L) on macroeconomic factors using historical data. Then apply the stressed factor values to the estimated model to project losses.

Let R_t be the portfolio return at time t and $\boldsymbol{X}_t = (X_{1,t}, \ldots, X_{k,t})^\top$ be a vector of macro factor changes. The model is:

$$R_t = \alpha + \sum_{i=1}^{k} \beta_i X_{i,t} + \varepsilon_t$$

The **stressed P&L** is:

$$\Delta\,\text{P\&L}_{\text{stress}} = V \cdot \left(\hat{\alpha} + \sum_{i=1}^{k} \hat{\beta}_i \, \Delta X_i^{\text{stress}} \right)$$

where V is the portfolio value and $\Delta X_i^{\text{stress}}$ is the change in macro factor i under the stress scenario.

Example 4.1 (Regression-based transmission)**.** Fit a quarterly regression of equity portfolio returns on macro factors using 40 quarters (10 years) of data:

$$R_t = \alpha + \beta_1 \Delta\text{GDP}_t + \beta_2 \Delta\text{Unemp}_t + \beta_3 \Delta\text{Spread}_t + \varepsilon_t$$

Estimated coefficients (representative):

Factor	$\hat{\beta}$	**Stressed** ΔX	$\hat{\beta} \times \Delta X$
ΔGDP (Y/Y, %)	$+2.8$	-6.0	-16.8%
ΔUnemployment (%)	-3.5	$+5.8$	-20.3%
ΔBBB Spread (bps/100)	-8.2	$+4.5$	-36.9%
Intercept $\hat{\alpha}$			$+1.2\%$
Total stressed return			$\mathbf{-72.8\%}$

Problem: the linear regression projects a -72.8% loss, which exceeds the 2008 actual equity decline of -57%. The model extrapolates linearly beyond the range of observed data, and the three factor effects may overlap (multicollinearity).

Mitigation: cap the projected loss at a historically calibrated maximum, use regularized regression (Ridge, Lasso), or use the satellite approach instead.

Warning

The regression approach is simple but fragile. Key pitfalls:

- **Multicollinearity**: GDP, unemployment, and credit spreads are highly correlated. OLS coefficients become unstable and can have wrong signs.
- **Linearity**: the true relationship between macro shocks and portfolio losses is almost certainly nonlinear, especially under extreme stress.
- **Extrapolation**: the stress scenario is, by design, outside the range of historical data used to estimate the model. Linear extrapolation can produce absurd results.
- **Structural breaks**: the sensitivity of the portfolio to macro factors may change over time as portfolio composition evolves.

4.2 Approach 2: Satellite Models (Bottom-Up)

Core Idea

Instead of one regression for the entire portfolio, build **separate models** for each component of portfolio loss, each driven by the relevant macro factors:

- **PD model**: $\text{PD}_t = f_1(\text{GDP}_t, \text{Unemp}_t, \ldots)$
- **LGD model**: $\text{LGD}_t = f_2(\text{HPI}_t, \text{Spread}_t, \ldots)$
- **Prepayment model**: $\text{CPR}_t = f_3(\text{Rate}_t, \text{HPI}_t, \ldots)$
- **Revenue model**: $\text{PPNR}_t = f_4(\text{Rate}_t, \text{GDP}_t, \ldots)$

Each satellite model is estimated independently, using data and features appropriate to the specific risk component. The stressed loss is assembled by feeding macro scenario paths into each model and aggregating.

The satellite approach is what large banks actually use for DFAST/CCAR submissions. It is more granular, more defensible, and more transparent than the top-down regression.

Example 4.2 (Satellite PD model)**.** A logistic regression for commercial loan default probability:

$$\ln\left(\frac{\mathrm{PD}_t}{1-\mathrm{PD}_t}\right) = \gamma_0 + \gamma_1\,\mathrm{GDP}_t + \gamma_2\,\mathrm{Unemp}_t + \gamma_3\,\mathrm{Spread}_t$$

Estimated on 80 quarters of data (20 years), with quarterly PD computed from the observed default rate on the bank's commercial loan portfolio.

Representative estimates: $\hat{\gamma}_0 = -4.2$, $\hat{\gamma}_1 = -0.15$, $\hat{\gamma}_2 = 0.22$, $\hat{\gamma}_3 = 0.35$.

Under the severely adverse scenario (GDP $= -6\%$, Unemp $= 10\%$, Spread $= +450$bps):

$$\mathrm{logit}(\mathrm{PD}_{\mathrm{stress}}) = -4.2 + (-0.15)(-6) + 0.22(10) + 0.35(4.5) = 0.075$$

$$\mathrm{PD}_{\mathrm{stress}} = \frac{1}{1+e^{-0.075}} \approx 51.9\%?$$

This is clearly too high — the issue is that the logistic model is extrapolating well beyond the training range. In practice, banks constrain PD projections using expert judgment, historical maximums, and model overlays. A more realistic stressed PD for a commercial loan portfolio would be 4–6%, achieved by calibrating the model with additional controls.

Key Insight

Both the regression and satellite approaches face the same fundamental challenge: stress scenarios are, **by construction**, outside the range of historical data. Any model — linear regression, logistic regression, machine learning — must extrapolate beyond its training domain. This is why stress testing always requires **expert judgment** as a complement to models. The model provides a disciplined starting point; judgment provides the guardrails.

5 Credit Migration Under Stress

For credit portfolios, losses arise not only from outright defaults but from **rating downgrades** that reduce the market value of bonds and loans. A transition matrix describes how borrowers migrate between rating categories over time.

5.1 The Transition Matrix

Definition 5.1 (Credit transition matrix)**.** A $K \times K$ **transition matrix** $\boldsymbol{P} = [p_{ij}]$ where p_{ij} is the probability that a borrower rated i at time t is rated j at time $t+1$. Rows sum to one: $\sum_{j=1}^{K} p_{ij} = 1$ for all i. The last column ($j = K$) represents default, which is an absorbing state.

Example 5.1 (One-year transition matrix (stylized))**.** A simplified 4-state matrix (A, BBB, BB, Default):

	A	**BBB**	**BB**	**Default**
A	91.5%	7.0%	1.2%	0.3%
BBB	4.0%	88.0%	6.5%	1.5%
BB	0.5%	6.0%	85.0%	8.5%
Default	0%	0%	0%	100%

Under normal conditions, an A-rated borrower has a 91.5% chance of remaining A, a 7.0% chance of being downgraded to BBB, and only a 0.3% chance of defaulting.

5.2 Stressing the Transition Matrix

Under economic stress, two things happen simultaneously:

1. **Downgrades accelerate**: the probability of moving to a lower rating increases.
2. **Upgrades slow**: the probability of moving to a higher rating decreases.

Two approaches to stress the matrix.

5.2.1 Approach 1: Multiplicative Stress Factors

Multiply the off-diagonal downgrade probabilities by a stress factor $s > 1$ and the off-diagonal upgrade probabilities by a factor $1/s$, then rescale rows to sum to one.

For a stress factor $s = 2.0$ applied to Example 5.1:

	A	**BBB**	**BB**	**Default**
A	83.0%	14.0%	2.4%	0.6%
BBB	2.0%	82.0%	13.0%	3.0%
BB	0.25%	3.0%	79.75%	17.0%

The A-rated default probability doubled from 0.3% to 0.6%; the BB default probability doubled from 8.5% to 17.0%.

5.2.2 Approach 2: Generator Matrix (Matrix Exponential)

Core Idea

A more rigorous approach uses the **generator matrix** $\boldsymbol{Q}$ such that $\boldsymbol{P} = \exp(\boldsymbol{Q})$ (matrix exponential). To stress the matrix, scale the generator:

$$\boldsymbol{P}_{\text{stress}} = \exp(s \cdot \boldsymbol{Q}), \quad s > 1$$

This is equivalent to "speeding up" the credit deterioration process by a factor of s. The resulting stressed matrix is guaranteed to be a valid transition matrix (all entries non-negative, rows sum to one) — unlike the multiplicative approach, which requires ad-hoc rescaling.

Formally, $\boldsymbol{Q} = \ln(\boldsymbol{P})$ is the matrix logarithm (when it exists with non-negative off-diagonal entries). The generator has the property that $q_{ij} \geq 0$ for $i \neq j$ and $q_{ii} = -\sum_{j \neq i} q_{ij}$ (rows sum to zero).

5.3 Worked Example: Investment-Grade Portfolio Under Stress

Example 5.2 (Stressed migration losses)**.** Consider a \$10 billion investment-grade bond portfolio: 60% rated A, 40% rated BBB. Under the stressed transition matrix ($s = 2.0$):

Rating distribution after one year:

- Originally A (\$6B): 83.0% stays A, 14.0% becomes BBB, 2.4% becomes BB, 0.6% defaults.
- Originally BBB (\$4B): 2.0% upgrades to A, 82.0% stays BBB, 13.0% becomes BB, 3.0% defaults.

Default losses (assuming LGD = 60%):

$$\text{Default loss} = (6 \times 0.006 + 4 \times 0.030) \times 0.60 = (0.036 + 0.120) \times 0.60 = \$93.6\text{M}$$

Mark-to-market migration losses: when a bond is downgraded, its spread widens and its market value falls. Assume the following spread impacts:

- A → BBB: spread widens by 50bps ⇒ price drop ≈ 2.5% (at 5Y duration).
- A → BB: spread widens by 200bps ⇒ price drop ≈ 10%.
- BBB → BB: spread widens by 150bps ⇒ price drop ≈ 7.5%.

Migration losses:

$$\begin{aligned} \text{A} \to \text{BBB}: &\quad 6 \times 0.14 \times 0.025 = \$21.0\text{M} \\ \text{A} \to \text{BB}: &\quad 6 \times 0.024 \times 0.10 = \$14.4\text{M} \\ \text{BBB} \to \text{BB}: &\quad 4 \times 0.13 \times 0.075 = \$39.0\text{M} \end{aligned}$$

Total stressed loss: \$93.6M + \$21.0M + \$14.4M + \$39.0M = \$168.0M (1.68% of portfolio value).

Under the unstressed matrix, the total loss would be approximately \$60M. Stress nearly triples the loss.

Interpretation

The majority of the stressed loss comes not from defaults but from **downgrades**. This is typical of investment-grade portfolios: defaults are rare even under stress, but the probability mass shifts downward through the rating spectrum, causing widespread spread widening and mark-to-market losses. For high-yield portfolios, the balance shifts toward default losses.

6 Reverse Stress Testing

Reverse stress testing inverts the standard question. Instead of "given a scenario, what is the loss?" it asks "given a loss threshold, what is the scenario?"

6.1 The Optimization Formulation

Definition 6.1 (Reverse stress test)**.** Let $\boldsymbol{\Delta X} = (\Delta X_1, \ldots, \Delta X_k)^\top$ be a vector of macro factor changes and $L(\boldsymbol{\Delta X})$ be the portfolio loss as a function of those changes. Let L^* be the loss threshold (e.g., the loss that would breach the minimum capital ratio).

The reverse stress test solves:

$$\min_{\boldsymbol{\Delta X}} \|\boldsymbol{\Delta X}\|_{\boldsymbol{\Sigma}^{-1}} \quad \text{subject to} \quad L(\boldsymbol{\Delta X}) \geq L^*$$

where $\|\boldsymbol{\Delta X}\|_{\boldsymbol{\Sigma}^{-1}} = \sqrt{\boldsymbol{\Delta X}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\Delta X}}$ is the Mahalanobis norm using the historical covariance matrix $\boldsymbol{\Sigma}$ of macro factor changes.

Core Idea

The Mahalanobis norm measures the "plausibility" of the scenario. A scenario where GDP falls by 10% *and* unemployment falls by 5% (contradictory) has a large Mahalanobis distance because such a combination has never been observed. A scenario where GDP falls by 6% and unemployment rises to 10% (consistent with historical recessions) has a smaller Mahalanobis distance.

Minimizing $\|\boldsymbol{\Delta X}\|_{\boldsymbol{\Sigma}^{-1}}$ subject to $L \geq L^*$ finds the **most plausible scenario** (closest to the mean, in a covariance-adjusted sense) that produces the target loss. This is the "least surprising" way to reach the break point.

6.2 Solution Approaches

6.2.1 Gradient-Based Optimization

If $L(\boldsymbol{\Delta X})$ is differentiable (e.g., when using the linear regression model from Section 4), the problem becomes:

$$\min_{\boldsymbol{\Delta X}} \boldsymbol{\Delta X}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\Delta X} \quad \text{s.t.} \quad \boldsymbol{\beta}^\top \boldsymbol{\Delta X} \leq -L^*/V$$

This is a quadratic program with a linear constraint — solvable in closed form via the KKT conditions:

$$\boldsymbol{\Delta X}^* = -\frac{L^*/V}{\boldsymbol{\beta}^\top \boldsymbol{\Sigma} \boldsymbol{\beta}} \boldsymbol{\Sigma} \boldsymbol{\beta}$$

The optimal scenario moves each factor in proportion to its beta times its covariance with other factors. Factors with high betas *and* high variance dominate the scenario.

6.2.2 Grid Search

When $L(\boldsymbol{\Delta X})$ is nonlinear (satellite models, option-like payoffs):

1. Define a grid of plausible factor changes for each macro variable.
2. Evaluate $L(\boldsymbol{\Delta X})$ at each grid point.
3. Among all grid points where $L \geq L^*$, select the one with the smallest Mahalanobis distance.

For $k = 5$ factors with 10 grid points each, this requires $10^5 = 100{,}000$ evaluations — feasible for fast loss functions, prohibitive for slow simulation-based models. Latin hypercube sampling or importance sampling can reduce the computational cost.

6.3 Worked Example: Finding the Break Point

Example 6.1 (Reverse stress test)**.** A bank has CET1 capital of \$5.0B, risk-weighted assets of \$55B, and a minimum CET1 ratio requirement of 4.5% (i.e., minimum capital of \$2.475B). Projected PPNR over the stress horizon: \$1.5B.

The loss threshold that would breach the minimum:

$$L^* = \text{Capital} + \text{PPNR} - \text{Minimum Capital} = 5.0 + 1.5 - 2.475 = \$4.025\text{B}$$

Using the linear model with $V = \$60\text{B}$, $\boldsymbol{\beta} = (2.8, -3.5, -8.2)^\top$ (from Example 4.1), and the historical covariance matrix $\boldsymbol{\Sigma}$:

	ΔGDP	ΔUnemp	ΔSpread
ΔGDP	4.0	−2.5	−1.2
ΔUnemp	−2.5	3.0	1.8
ΔSpread	−1.2	1.8	2.5

Solving the KKT conditions yields the critical scenario: $\Delta\text{GDP}^* \approx -4.1\%$, $\Delta\text{Unemp}^* \approx +4.2\%$, $\Delta\text{Spread}^* \approx +310\text{bps}$.

This is a scenario less severe than the DFAST severely adverse case, which means the bank's capital buffer above the minimum is thin — a moderate recession, not a deep one, would be sufficient to breach the floor.

Warning

Reverse stress testing often reveals uncomfortable truths. If the "break point" scenario is a moderate recession rather than an extreme crisis, the institution's capital position is weaker than the forward stress tests alone might suggest. This is precisely why reverse stress testing is valuable — and why some institutions resist implementing it rigorously.

7 Historical Scenario Replay

Historical scenario replay is the most intuitive form of stress testing: take the factor moves that occurred during a past crisis and apply them to the current portfolio.

7.1 The Method

1. **Select a crisis period** and define the stress window (e.g., September 15, 2008 to March 9, 2009 for the GFC).

2. **Extract factor changes**: compute the cumulative change in each risk factor over the stress window.

3. **Revalue the portfolio**: apply the stressed factor changes to the current portfolio positions using either full revaluation or sensitivity-based (delta-gamma) approximation.

4. **Compute the loss**: the difference between current portfolio value and the stressed portfolio value.

7.2 Three Key Historical Scenarios

Factor	GFC 2008	COVID 2020	SVB 2023
S&P 500	−57% (peak-to-trough)	−34% in 23 days	−5% (broad); bank stocks −60%
VIX	14 → 80	14 → 82	18 → 31
10Y Treasury	−300bps	−120bps	+400bps (2022–23 hiking cycle)
BBB Spread	+400bps	+350bps	+80bps (banks +250bps)
House Prices	−27% (Case-Shiller, over 2 years)	+5% (no decline)	flat
Unemployment	5% → 10%	3.5% → 14.7% (April 2020)	3.4% (stable)
Duration	Sep 2008 – Mar 2009 (6 months)	Feb 19 – Mar 23 (23 days)	Mar 8–12 (3 days for SVB)

Table 3: Key factor moves during three recent crises. Each crisis has a distinct "signature" — different factors dominate in each.

7.3 Worked Example: Applying the GFC Scenario

Example 7.1 (GFC replay on a current portfolio)**.** Consider a \$100M portfolio: 60% equities, 25% investment-grade bonds (5Y duration), 15% high-yield bonds (4Y duration).

Apply the GFC factor moves:

- **Equities** (−57%): $\$60\text{M} \times (-0.57) = -\$34.2\text{M}$.
- **IG bonds**: spread widens by 400bps on a 5Y duration portfolio. Price change $\approx -5 \times 4.0\% = -20\%$. Loss: $\$25\text{M} \times 0.20 = -\$5.0\text{M}$. Partially offset by rates falling 300bps: $+5 \times 3.0\% = +15\%$. Gain: $\$25\text{M} \times 0.15 = +\$3.75\text{M}$. Net IG loss: $-\$1.25\text{M}$.
- **HY bonds**: spread widens by 800bps (HY spreads moved more than IG). Price change $\approx -4 \times 8.0\% = -32\%$. Rate offset: $+4 \times 3.0\% = +12\%$. Net: -20%. Loss: $\$15\text{M} \times 0.20 = -\$3.0\text{M}$.

Total GFC scenario loss: $\$34.2\text{M} + \$1.25\text{M} + \$3.0\text{M} = \$38.45\text{M}$ (38.5% of portfolio value).

Key insight: the equity allocation dominates the loss. The bond portfolio partially benefits from the Treasury rally (flight to quality), which offsets some of the spread widening on IG bonds. But HY bonds get hit from both sides during a GFC-type event.

Example 7.2 (COVID replay)**.** Same portfolio, COVID March 2020 moves:

- **Equities** (−34%): $-\$20.4\text{M}$.
- **IG bonds**: spread +350bps, rates −120bps. Net spread impact: $-5 \times 3.5\% + 5 \times 1.2\% = -11.5\%$. Loss: $\$25\text{M} \times 0.115 = -\$2.875\text{M}$.
- **HY bonds**: spread +700bps, rates −120bps. Net: $-4 \times 7.0\% + 4 \times 1.2\% = -23.2\%$. Loss: $\$15\text{M} \times 0.232 = -\$3.48\text{M}$.

Total COVID scenario loss: $\$20.4 + \$2.875 + \$3.48 = \$26.76\text{M}$ (26.8%).

Example 7.3 (SVB/rate shock replay)**.** The SVB 2023 scenario is fundamentally different: it is a **rate shock**, not a credit shock. Equities fall modestly, but the damage comes from rising rates destroying the value of long-duration fixed-income holdings.

Same portfolio, SVB-style moves (rates +400bps, equities −5%, bank stocks −60%):

- **Equities** (−5% broad market): $-\$3.0\text{M}$.
- **IG bonds**: rates +400bps, spread +80bps. Price change: $-5 \times (4.0\% + 0.8\%) = -24\%$. Loss: $\$25\text{M} \times 0.24 = -\$6.0\text{M}$.
- **HY bonds**: rates +400bps, spread +80bps. Price change: $-4 \times 4.8\% = -19.2\%$. Loss: $\$15\text{M} \times 0.192 = -\$2.88\text{M}$.

Total SVB scenario loss: $\$3.0 + \$6.0 + \$2.88 = \$11.88\text{M}$ (11.9%).

The SVB scenario is less severe in total but exposes a different vulnerability: the bond portfolio suffers *without* the offsetting rate decline that cushions spread-driven losses in the GFC and COVID scenarios. In a rising-rate environment, there is no flight-to-quality offset.

Key Insight

Each historical crisis has a distinct **risk factor signature**:

- **GFC 2008**: massive equity decline + credit spread blowout, partially offset by rate decline.
- **COVID 2020**: rapid equity decline + credit spread blowout + rate decline, compressed into 23 days.
- **SVB 2023**: rate shock without credit spread relief, concentrated in bank-exposed sectors.

Running all three reveals different portfolio vulnerabilities. A portfolio that survives the GFC replay may fail the SVB replay, and vice versa. This is why multiple historical scenarios are essential.

Interpretation

Notice that the GFC scenario produces the largest absolute loss ($38.5M), but the COVID scenario produces the largest loss **per unit of time** ($26.8M in 23 days vs. $38.5M in 6 months). The speed of the COVID drawdown meant that portfolios had virtually no time to rebalance. The SVB scenario, while producing the smallest loss overall, is the most dangerous for portfolios with significant duration exposure and no hedges — exactly the position that caused Silicon Valley Bank to fail.

8 References

The following texts provide the regulatory, theoretical, and practical foundations for stress testing in financial risk management.

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[7] **Rebonato, R.** (2010). *Coherent Stress Testing: A Bayesian Approach to the Analysis of Financial Stress.* Wiley. Bayesian framework for constructing internally consistent stress scenarios.

[8] **Glasserman, P., Kang, C. & Kang, W.** (2015). "Stress scenario selection by empirical likelihood." *Quantitative Finance*, 15(1), 25–41. Systematic approach to selecting worst-case scenarios from historical data.